Biological Transitions as Multi-Agent Realisations of the Generative Operator Pipeline in Topographical Orthogonal Generative Theory (TO/TOGT)

Pablo Nogueira Grossi

March 2026

Abstract

We model collective biological transitions—including neural oscillations, HPA-axis stress response, circadian regulation, immune adaptation, and protein conformational change—as multi-agent realisations of the generative operator pipeline $G = U \circ F \circ K \circ C$ introduced in Topographical Orthogonal Generative Theory (TO/TOGT). Each agent undergoes local compression, curvature intensification, loss of injectivity, and stabilisation. The collective fixed points, contraction properties, and saturated pitchfork bifurcation behaviour follow directly from the fixed-point theory, contraction-mapping results, and algebraic separation theorem proved in the author's earlier work (HAL hal-05555216v1). The specific dm3 numerical invariants ($T^* = 2\pi$, $\mu_{\max} = -2$, $\tau = 2$) are convenient modelling choices matching observed biological scales; they are not derived theorems of the operator algebra. No numerical simulations or unverified biological claims are made.

MSC codes: 37C25, 37D10, 37G10, 47H10, 92B20, 92C20.

1 Relation to Prior Work

This article applies the mathematical results established in the author's earlier preprint *Principia Orthogona, Volume One: The Mathematics of Generative Transitions* (HAL hal-05555216v1, Zenodo 19 March 2026). There the composite operator pipeline

$$G = U \circ F \circ K \circ C$$

was introduced together with complete proofs of fixed-point existence and uniqueness, global contraction, saturated pitchfork bifurcation, and the algebraic trace-bound/separation theorem.

The geometric realisation of the operator pipeline on contact manifolds was developed in *Principia Orthogona, Volume Two: Contact Realisation of Generative Transitions* (HAL hal-05559997v1 and hal-05565024v1, both deposited 24 March 2026).

The present paper demonstrates a multi-agent biological application of the same operator algebra. It is directly related to the following Zenodo deposits by the author:

- Zenodo record 19162013: `https://zenodo.org/records/19162013`
- Zenodo record 19122168: `https://zenodo.org/records/19122168`
- Zenodo record 19117400: `https://zenodo.org/records/19117400`

All mathematical claims in this paper follow strictly from previously proved results.

2 Operator Pipeline in Biological Systems

Definition 2.1 (Biological Agent Dynamics)**.** *Let each biological agent (neuron, endocrine cell, immune cell, protein domain) follow a trajectory $x_i(t)$ in a Riemannian manifold (M, g). The collective biological state is the tuple $(x_1, \dots, x_N)$. Each agent undergoes a generative transition governed by the composite operator*

$$G = U \circ F \circ K \circ C,$$

where:

- *C: local orthogonal compression (neighbourhood averaging),*
- *K: curvature intensification (alignment and clipping),*
- *F: loss of injectivity (folding of local information),*
- *U: stabilisation (global coherence).*

The contraction-mapping theorem of Volume One ensures that for sufficiently small coupling between agents, the collective system converges to a unique fixed point.

3 Fruit Fly Connectome as Multi-Orbit System

The FlyWire connectome provides a concrete multi-agent biological system. Each neuron trajectory is modelled as an identity orbit $\mathcal{O}_i$. The full connectome is the multi-orbit system

$$\mathcal{S} = \{\mathcal{O}_1, \dots, \mathcal{O}_N\}.$$

The contraction results of Volume One imply that under small inter-neuron coupling, the system converges to a collective fixed point preserving local connectivity while achieving global coherence. This is a direct biological instance of multi-orbit identity stabilisation.

4 HPA-Axis Stress Response as Saturated Pitchfork

The hypothalamic-pituitary-adrenal (HPA) axis exhibits threshold behaviour under acute stress. This is modelled by the saturated pitchfork bifurcation proved in Volume One:

$$x' = \lambda x - cx^3,$$

with clipping enforcing bounded response. At critical coupling $|\alpha| = 0.5$, the system transitions between low-stress and high-stress attractors. The clipping function κ prevents runaway amplification and ensures recovery.

5 Discussion

The TO/TOGT operator pipeline provides a unified dynamical-systems language for describing collective biological transitions. All mathematical claims in this paper follow directly from rigorously proved theorems in HAL hal-05555216v1 and the contact-geometric realisation in HAL hal-05559997v1 / hal-05565024v1. The specific dm3 numerical invariants are modelling choices matching biological scales; they are not derived from the operator algebra.

Future work will focus on explicit multi-agent simulations and experimental falsification of predicted bifurcation thresholds.